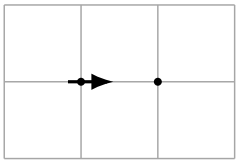
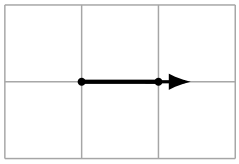
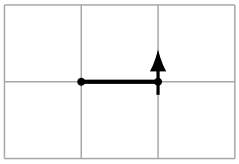
Use a slim 10 mm paper arrow. One move follows an edge to the next dot. Then turn left, turn right, or go straight. The arrow shows which way you face. One partner moves; the other checks. Swap roles.



Start

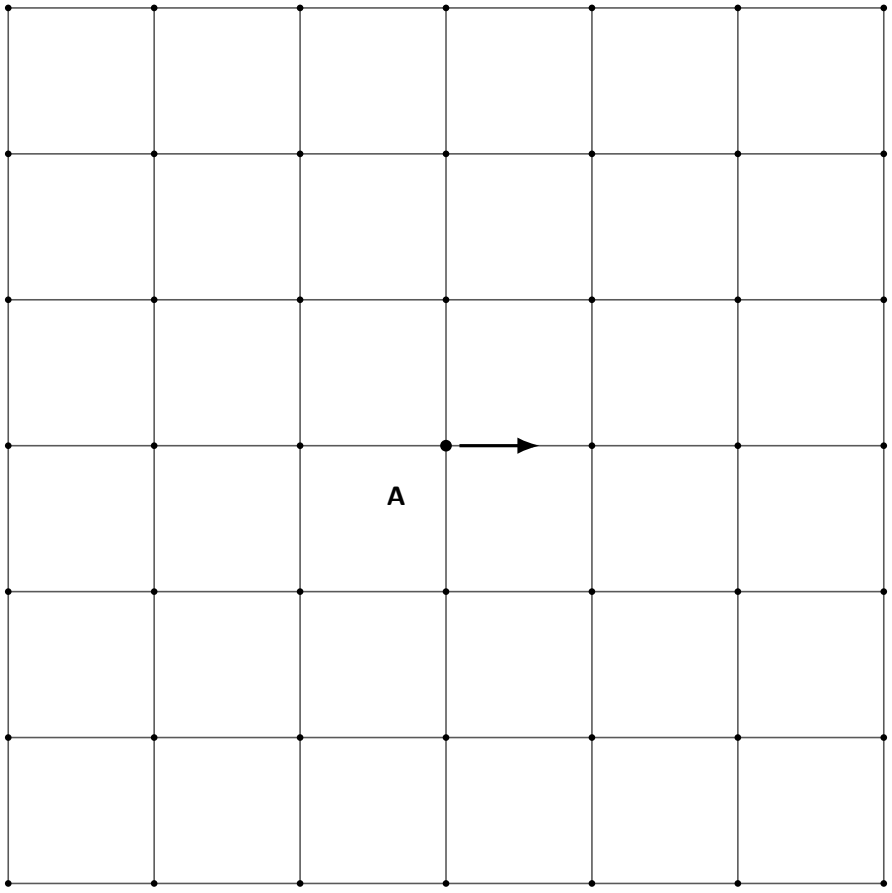


Move one edge



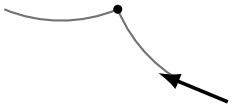
Turn left

Problem 1: Start at A, facing along the arrow. Find routes of 4, 6, and 8 moves that return to A facing the same way. Never turn around.

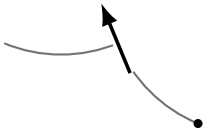


Each octagon has eight right-angle corners. The rooms are all the same size in this geometry. The map makes straight streets look curved.

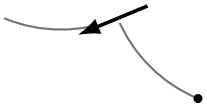
Keep the arrow along the road as it bends. Turn left or right from the arrival direction.



Start along the road

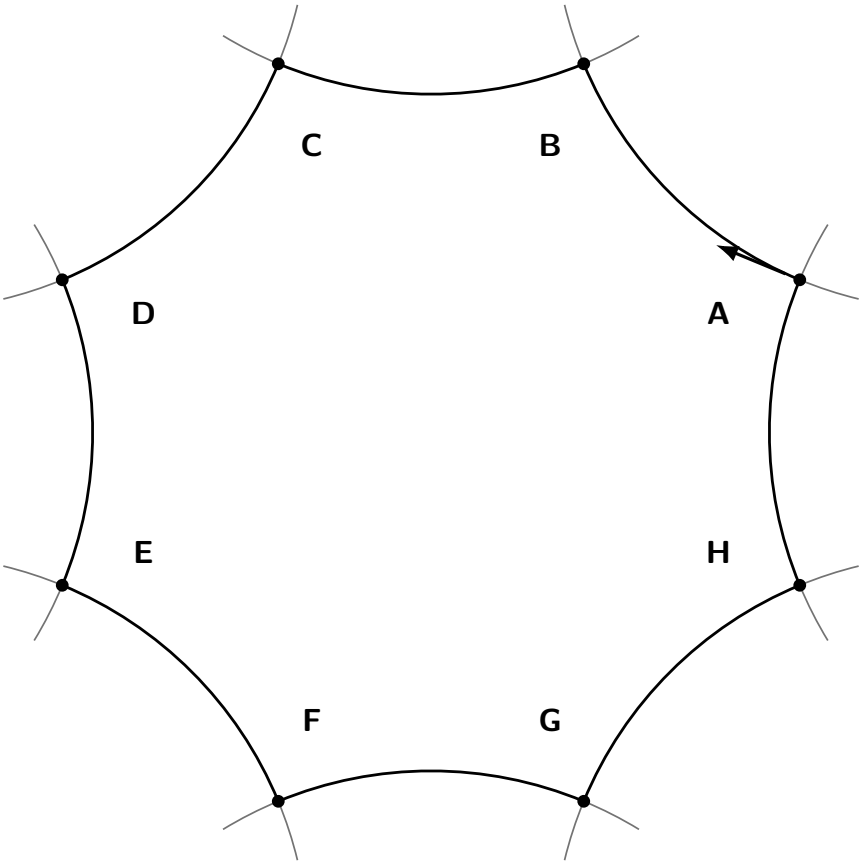


Arrive along the road



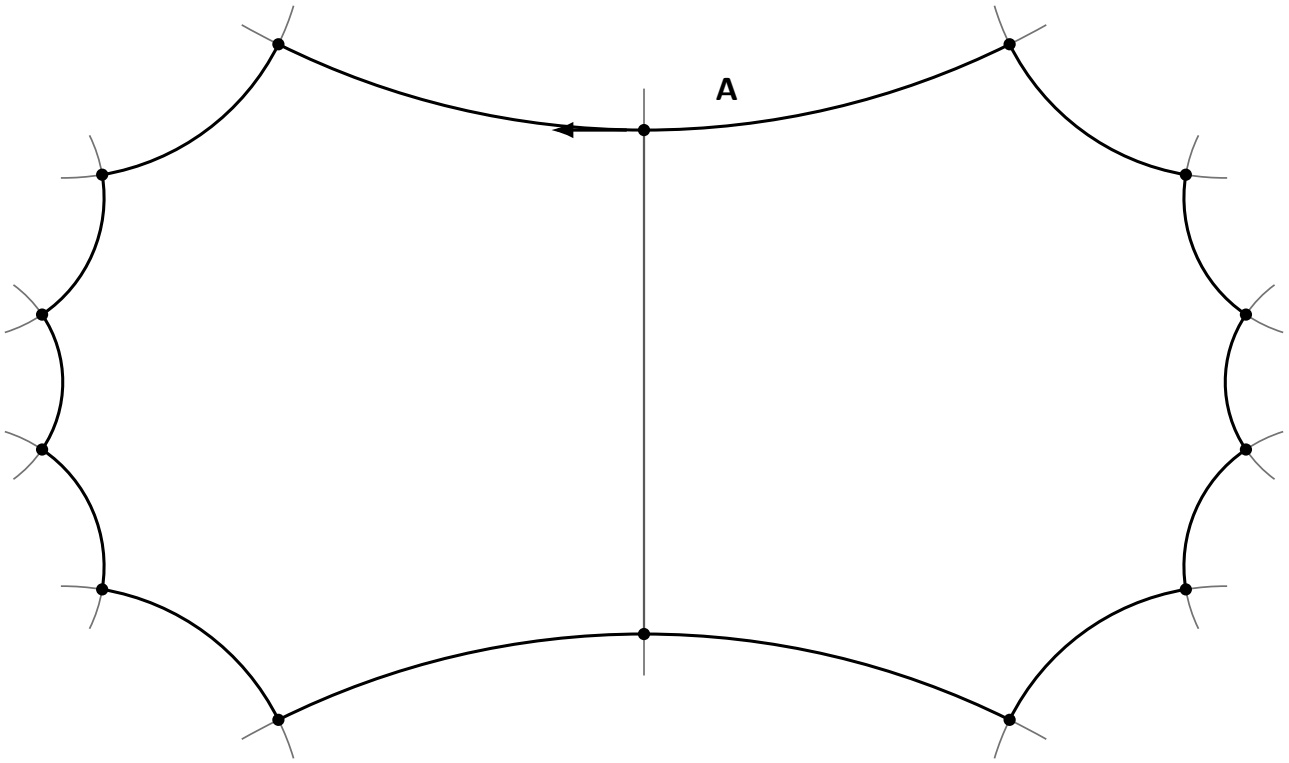
Turn left

Problem 2: Start at A, facing along the road toward B. Move one edge and turn left, over and over. Mark where you are after 4 moves. How many moves make the first return to your starting place, facing the same way? Start toward H and try only right turns.

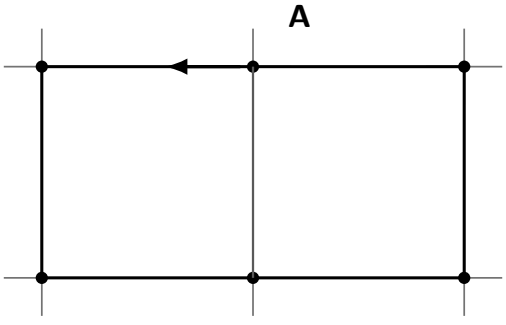


Left turns: _____ moves Right turns: _____ moves

Problem 3: Walk once around the outside of both rooms, starting at A along the arrow. Keep both rooms on your left. Count only the turns at dots. Compare the two octagons with the two squares.



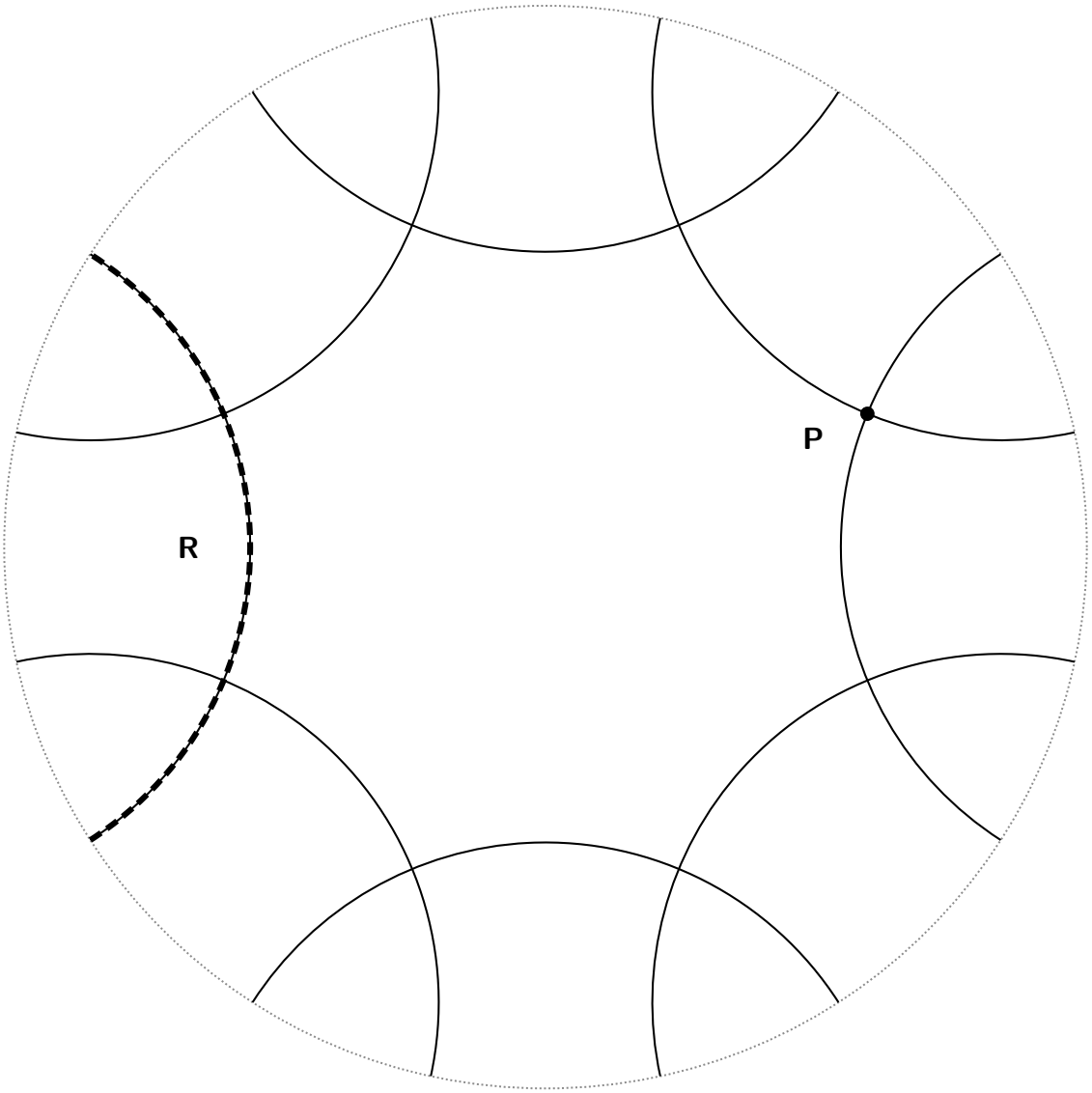
Around two octagons: _____ left turns



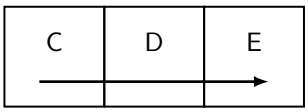
Around two squares: _____ left turns

A complete street goes straight through every crossing, taking the opposite branch. Streets continue forever. The dotted circle is the edge of the map; no number of moves reaches it. Only some streets are drawn.

Problem 4: Trace every drawn complete street through P in both directions. How many miss the heavy dashed street R? Can two different ordinary straight lines through one point both miss a third line? The ordinary lines also continue forever in both directions. Draw or explain below.



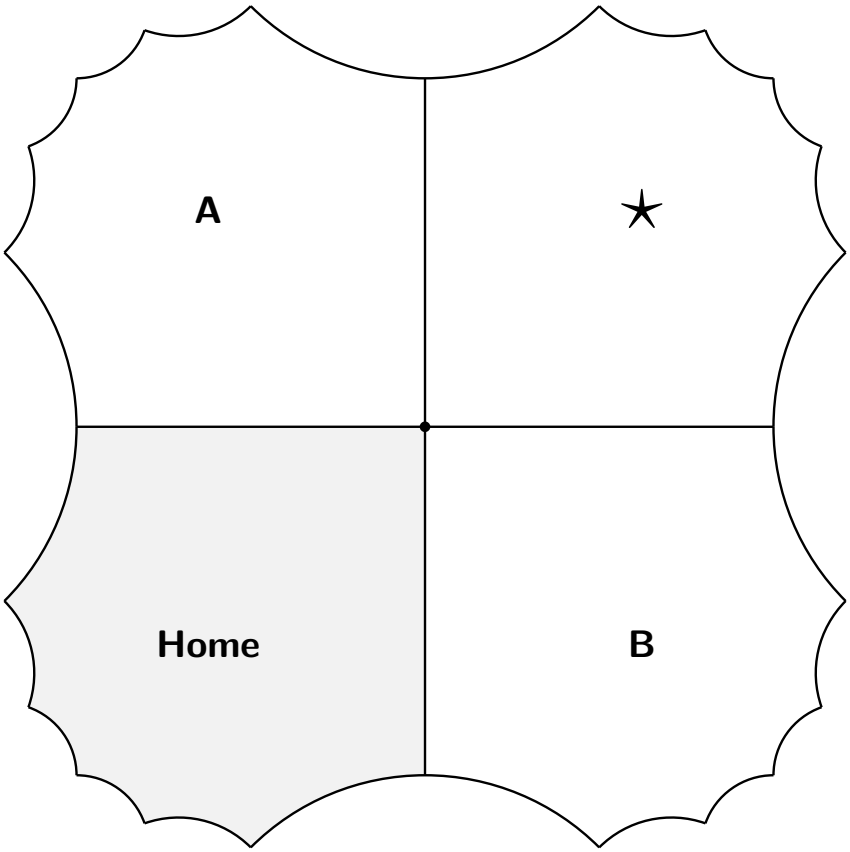
For room routes, put a coin inside a room. A door crossing moves it across one shared side into the next room. Crossing a corner is not allowed. Write the rooms in the order you visit them.



$C \rightarrow D \rightarrow E$

Two door crossings

Problem 5: This is another view of the same geometry, showing four rooms around one corner. Find every room route from Home to $\star$ using exactly 2 door crossings. Then find every route using exactly 4 door crossings. Stay in these four rooms. You may visit a room or use a door more than once.



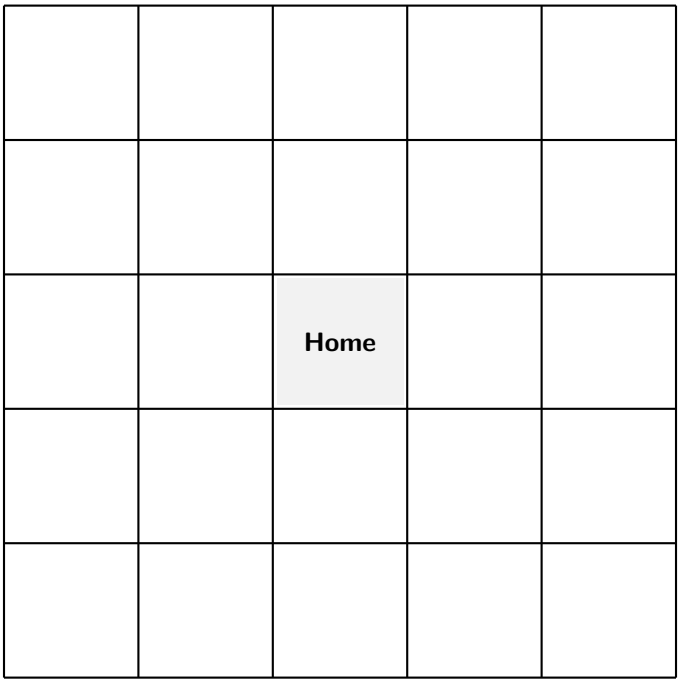
Keep the route sheet beside this board.

Problem 5 (continued)

2 door crossings

4 door crossings

Problem 6: On the octagon floor every room has 8 doors. How many rooms can you reach from Home in 2 door crossings but not fewer? How many on this square floor, where every room has 4 doors?



Octagon floor: _____ rooms Square floor: _____ rooms